

SS2P2: A Stable State-Space Point Process for Limit-Order-Book Simulation

Anonymous submission

Abstract

Simulating limit-order-book (LOB) order flow is a core generative task for market-making world models, agent training, and counterfactual analysis. State-Space Point Process (S2P2) predicts the next LOB event accurately, but in closed loop its learned self-excitation drifts or explodes. We propose SS2P2 (Stable State-Space Point Process), which keeps the state-space backbone of S2P2 and replaces only the output heads: a softmin-bounded rate head whose intensity has a hard closed-form ceiling (an exact dominating rate for Ogata thinning) and a floor of exactly zero, and a rate-neutral softmax mark head that can be arbitrarily deep without perturbing the total rate. On three Coinbase assets, against five baselines with three training seeds each and verified rate calibration for every model, SS2P2 is the strongest predictor on two of three assets while remaining verifiably calibratable in closed loop with competitive simulated structure. SS2P2 trains with an exact parallel scan and simulates in $O(1)$ per event by exact Ogata thinning. These results establish SS2P2 as a likelihood-trained model that both predicts accurately and simulates stably, a practical foundation for market-making world models.

Introduction

The limit order book (LOB) is the core mechanism of electronic markets: a continuous-time stream of order placements, cancellations, and trades whose fine structure drives price discovery and liquidity. Two tasks are asked of a generative model of this stream. The first is *prediction*: given the event history, score the time and type of the next event. The second is *simulation*: roll the model forward freely, feeding its own samples back as history, and reproduce the stylized facts of real order flow, such as calibrated event rates, burst-lull clustering, and heavy-tailed returns (Cont 2001). Simulation is the harder task and, for downstream use (market-making world models, agent training, counterfactual analysis), the more valuable one (Li et al. 2025; Nagy et al. 2025).

Neural MTPPs (Du et al. 2016; Mei and Eisner 2017; Shchur et al. 2021) excel at prediction, and the continuous-time state-space model S2P2 (Chang et al. 2025; Shi and Cartlidge 2022) is the current state of the art: it holds the best held-out likelihood among neural TPPs. It is therefore the natural candidate for an LOB simulator. But its predictive power has only ever been demonstrated teacher-forced. The S2P2 paper never rolls the model out, and we find that when it is rolled out, the state-of-the-art predictor is not a

usable simulator. Two obstacles are responsible. *First, instability*: the self-excitation that wins the one-step likelihood becomes a liability in closed loop, because each sampled event raises the intensity, which produces more events, and with no mechanism bounding the rate the roll-out drifts hot or explodes. *Second, uncalibratability*: the standard patch, rescaling the intensity until the simulated rate matches reality, turns out to be unavailable exactly where it is needed most. We find trained S2P2 checkpoints whose closed-loop rate is *near-discontinuous* in the rescaling constant, so that no constant rescaling exists. On one of our three assets, every S2P2 checkpoint fails, leaving no usable simulator at all. The gap, then, is concrete: the model with the best prediction power cannot simulate stably, while simple stable models simulate safely but predict poorly.

SS2P2 changes only the output heads of S2P2. The stacked latent-linear-Hawkes state-space backbone is kept verbatim, with its diagonal zero-order-hold dynamics, parallel-scan computability, and LayerNorm’d stack output h_t . Everything after h_t is replaced. Because the backbone is identical, differences against the unbounded parent are attributable to the heads. Our contributions:

1. **The SS2P2 head pair**: a softmin-bounded rate head, whose intensity has a hard closed-form ceiling (an exact dominating rate for Ogata thinning) and an exactly-zero floor with unit log-intensity gradient in quiet periods, times a rate-neutral soft-max mark head that cannot change the total rate however deep its network. Together they turn S2P2 into a provably stable simulator.
2. **Stable training and simulation**. Stateful (TBPTT) training walks contiguous stretches of the stream so the model never trains in a cold-start transient, and closed-loop roll-outs carry the packed decoder state at $O(1)$ cost per event, mathematically identical to encoding the full history and free of the sliding window’s memory ceiling. Training and simulation see the same regime.
3. **Verified, exact post-hoc rate calibration**. For the factorized head the rate is linear in a learnable scale, the ceiling scales with it, and marks are untouched, so calibration is a one-dimensional root-find that keeps the guarantee intact. A mark-preserving variant applies to every baseline, with full-scale verification. Under this test, calibratability becomes a measurable model property: the

unbounded parent frequently fails with a critical-point signature while the bounded head always verifies.

Background and Related Work

Marked temporal point processes. An MTPP models a sequence $\mathcal{H}_T = \{(t_i, e_i)\}_{i=1}^{N_T}$ of events with times t_i and types $e_i \in \mathcal{E}$, a finite set of atomic event types, through per-type conditional intensities $\lambda_e(t \mid \mathcal{H}_t)$. With total intensity $\lambda(t) = \sum_e \lambda_e(t)$ and mark distribution $p(e \mid t) = \lambda_e(t)/\lambda(t)$, the log-likelihood is

$$\log \mathcal{L} = \sum_i \left[\log \lambda(t_i) + \log p(e_i \mid t_i) \right] - \int_0^T \lambda(u \mid \mathcal{H}_u) du,$$

the integral being the compensator $\Lambda(t) = \int_0^t \lambda(u \mid \mathcal{H}_u) du$. Sampling uses Ogata thinning, which requires a dominating rate $\bar{\lambda} \geq \lambda(t)$. A model whose intensity has no computable upper bound must resort to heuristic, bias-prone dominating-rate estimates.

The S2P2 backbone. S2P2 (Chang et al. 2025; Shi and Cartlidge 2022) is a state-space point process: L stacked latent-linear-Hawkes/SSM layers with diagonal state, ZOH-discretized between events and computable by parallel scan, $x_{t'}^{(\ell)} = \bar{A}^{(\ell)} x_t^{(\ell)} + (\bar{A}^{(\ell)} - I) \tilde{B}^{(\ell)} h^{(\ell-1)} + \tilde{E}^{(\ell)} \alpha$ at an event], with a LayerNorm’d residual readout $h^{(\ell)} = \text{LayerNorm}(\text{GELU}(y^{(\ell)}) + h^{(\ell-1)})$ producing the continuous-time hidden representation $h_t := h^{(L)}(t) \in \mathbb{R}^H$. The published model reads the rate from h_t through a projection-and-softplus head, $\lambda = \text{softplus}(w^\top h_t + b)$, unbounded above and with no learnable scale. Marks couple to the same rate field (our benchmark implementation splits h_t into disjoint halves for the rate and mark heads). SS2P2 keeps everything up to and including h_t and changes only the two heads.

Neural TPPs for event streams. Neural MTPPs learn history-dependent intensities with recurrent (Du et al. 2016; Mei and Eisner 2017), attention-based (Zhang et al. 2020; Zuo et al. 2020), jump-diffusion (Zhang et al. 2024), and continuous-time state-space (Chang et al. 2025) encoders (survey: Shchur et al. 2021). Shi and Cartlidge (2022) specialize neural Hawkes models to LOB streams and evaluate both prediction and simulation. The state-space line, however, is evaluated almost exclusively teacher-forced: Chang et al. (2025) report state-of-the-art held-out likelihood but never roll the model out. Our contribution is orthogonal: we keep the backbone verbatim and change only what is read out of it.

Hawkes modeling of order flow. Parametric Hawkes processes are the classical LOB event model (Hawkes 1971, 2018; Jain et al. 2024a,b), with state-dependent extensions (Morariu-Patrichi and Pakkanen 2022): exact stationarity certificates, limited expressiveness. SS2P2 takes the complementary route, full neural expressiveness with stability restored by a closed-form bound. In the neural TPP literature such bounds appear mainly as computational devices for thinning. Here the bound is a modeling decision with an asymmetric requirement (hard ceiling, exactly-zero floor).

Generative LOB simulators. Learned order-flow generators span GANs (Li et al. 2020; Coletta et al. 2021, 2022), token-level autoregressive state-space models (Nagy et al. 2023), diffusion-guided agents (Huang et al. 2026), and foundation-model-scale transformers (Li et al. 2025; Sood et al. 2026). Agent-based simulators (Byrd, Hybinette, and Balch 2020; Frey et al. 2023) remain the RL workhorse. These systems are scored on distributional realism: stylized facts (Cont 2001; Vyetrenko et al. 2020) and the KS/Wasserstein suites of LOB-Bench (Nagy et al. 2025), whose authors report error accumulation over long roll-outs. SS2P2 differs on two counts: one likelihood-based model is held to *both* prediction and closed-loop realism, and the simulated rate is calibrated and *verified*, a closed-loop stability certificate that no generative simulator above carries.

Methods

To apply point process modeling to high-frequency limit order book (LOB) data, we represent market dynamics as a sequence of **events** (pipeline overview in Figure 1). We collect the data, construct events, and train a Stable State-Space Point Process (SS2P2) model to learn the underlying dynamics.

Data collector

We obtain both trade and Level-2 limit order book data from Kaiko (Kaiko 2025), an institutional provider of digital-asset market data, and operate the data collector in partnership with Cloudburst Technologies (Cloudburst 2025), a firm specializing in detecting and mitigating market manipulation. Each update record contains a precise timestamp, the update type (snapshot or liquidity update), and the corresponding price–volume information for bid and ask levels. Complementary trade data are retrieved via Kaiko’s API and temporally aligned with the order book stream. This synchronization of trades and book updates, in both time and volume, is what the event constructor below relies on to classify every observed book change unambiguously.

Event constructor

To apply point process modeling to high-frequency limit order book (LOB) data, we represent market dynamics as a sequence of **events**. We consider the top $K = 10$ levels on each side (bid and ask).

Atomic Events. We define a finite set $\mathcal{E}$ of *atomic event types*, each denoted $e = (c, s, \delta)$, where:

- $c \in \{\text{LO}, \text{MO}, \text{CO}, \text{IS}\}$ is the *event class*:
 - **Limit Order (LO)**: liquidity provision at one of the top- K levels.
 - **Cancel Order (CO)**: liquidity withdrawal at one of the top- K levels.
 - **Market Order (MO)**: liquidity consumption at the best quote.
 - **Inside-Spread Order (IS)**: a new order placed inside the spread, improving the prevailing best quote.
- $s \in \{b, a\}$ is the *side* (bid or ask).

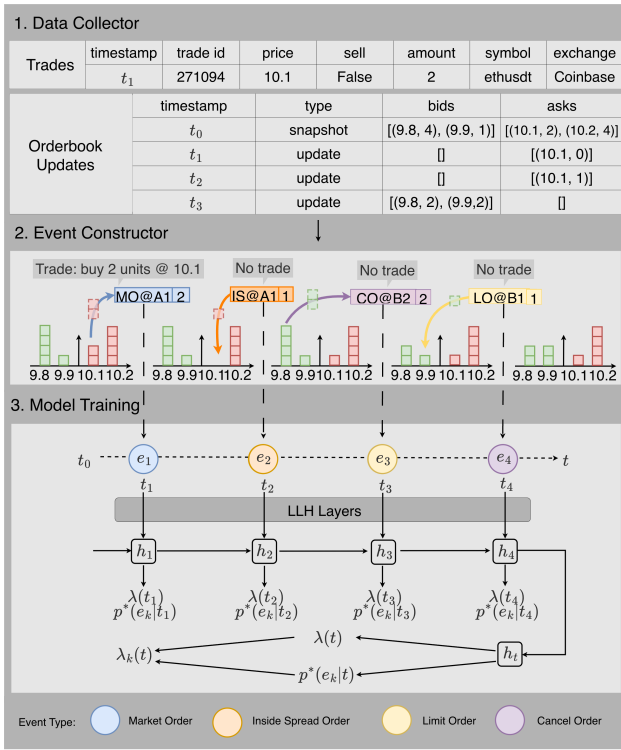


Figure 1: Pipeline. (1) *Data collector*: synchronized trade records and Level-2 order book updates. (2) *Event constructor*: consecutive book snapshots are differenced and aligned with trades to emit atomic typed events. (3) *Model training*: the event sequence drives the state-space backbone (LLH layers) whose continuous-time hidden representation h_t decodes the bounded total rate $\lambda(t)$ and the rate-neutral mark distribution $p(e | t)$, giving per-type intensities $\lambda_e(t) = \lambda(t)p(e | t)$.

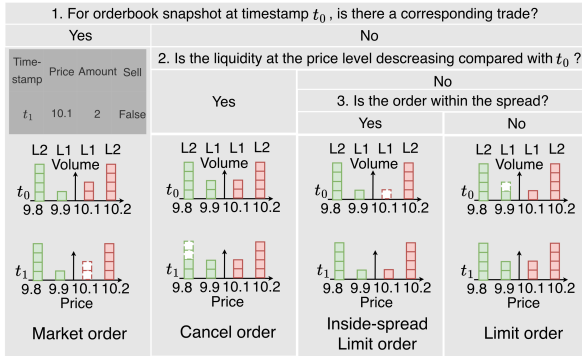


Figure 2: Event construction for the four order classes through three questions, each a decision-tree classification applied to synchronized Level-2 book snapshots and trade records at timestamps $t_0 \rightarrow t_1$.

- δ is the *level index*:
 - For LO/CO, $\delta \in \{1, \dots, K\}$ indicates distance from

the best quote ($\delta = 1 = \text{best bid/ask}$, $\delta = 10 = \text{tenth level}$).

- For MO, $\delta = 1$ always, since market orders execute against the current best quote.
- For IS, $\delta \in \{1, \dots, K\}$ denotes the number of ticks by which the new quote improves the spread (e.g., $IS_{b,2} = \text{new bid order two ticks inside the spread}$).

Event Construction from LOB and Trade Data. As illustrated in Figure 2, each atomic order event is inferred by differencing consecutive Level-2 order book snapshots and aligning them with trade records at identical timestamps. Because trades and book updates are perfectly synchronized in both time and volume, every observed change can be classified without ambiguity through a three-step decision tree. First, if the best-quote liquidity decreases and a trade is recorded at the same price, the change corresponds to a **Market Order**—a direct execution against standing quotes. Second, if liquidity decreases at any level without a matching trade, the reduction represents a **Cancel Order**, reflecting liquidity withdrawal. Finally, when liquidity increases, the classification depends on the price position: if the new order appears inside the spread, it forms an **Inside-Spread Order**; otherwise, it is a **Limit Order** placed within the existing book depth. This rule set ensures exhaustive mapping from raw order book and trade data into atomic order events, enabling a consistent foundation for downstream temporal modeling.

Stable State-Space Point Process (SS2P2)

SS2P2 factors the per-type intensity into a bounded total rate and a rate-neutral mark distribution — the ground-intensity–mark factorization of set-valued MTPPs (Chang, Boyd, and Smyth 2024) — both decoded from the *same* continuous-time hidden representation h_t :

$$\lambda_e(t) = \lambda(t) \cdot p(e | t), \quad \sum_{e \in \mathcal{E}} \lambda_e(t) = \lambda(t),$$

the second identity holding because $\sum_e p(e | t) = 1$. The design principle is to put nonlinearity where it cannot perturb the rate.

Softmin-bounded rate head

For S2P2, the closed-loop simulation fails through a feedback loop: each sampled event raises the intensity, a higher intensity produces events faster, and with an unbounded head the roll-out drifts hot or explodes. The rate head is built to break this loop. From the backbone’s continuous-time hidden representation h_t (LayerNorm’d, hence bounded in scale), a gate produces a bounded state g , which is read out through a smooth one-sided cap with fixed constant $z_{\max}$:

$$\begin{aligned} o &= \sigma(W_o h_t + b_o) \in (0, 1)^H, \\ g &= o \odot \tanh(h_t) \in (-1, 1)^H, \\ z_{\text{raw}} &= w^\top g + b \quad (\text{unconstrained}), \\ z &= z_{\max} - \text{softplus}(z_{\max} - z_{\text{raw}}), \\ \lambda(t) &= \varsigma \cdot \text{softplus}(z) \end{aligned}$$

with a learnable positive scale ς initialized so the baseline rate matches a target rate ($g \rightarrow 0 \Rightarrow \lambda \approx \varsigma \ln 2$). Above makes four deliberate design choices:

- **Bounded input.** The readout sees $g \in (-1, 1)^H$, not the raw latent state: however far the closed loop drives h_t — including states training never visited — an extreme state can change the direction of g but not its magnitude.
- **Hard ceiling.** Since $\text{softplus}(x) > 0$ for every x , $z \leq z_{\max}$ holds identically, so $\lambda(t) \leq \varsigma \cdot \text{softplus}(z_{\max})$ for every state and every parameter setting — no training outcome can remove the bound. The worst the roll-out can ever do is behave like a homogeneous Poisson process at the ceiling, and the ceiling doubles as an *exact* dominating rate for Ogata thinning.
- **Vanishing gain.** Differentiating the cap gives $\partial z / \partial z_{\text{raw}} = \sigma(z_{\max} - z_{\text{raw}})$: as a burst pushes the intensity toward the ceiling, the head becomes exponentially *less* sensitive to further state excursions, so the feedback loop loses gain exactly when it is most dangerous. This is what makes the closed-loop rate respond smoothly to rescaling — and hence what makes the model calibratable.
- **Open floor.** The cap is one-sided: as $z_{\text{raw}} \rightarrow -\infty$, $z \rightarrow z_{\text{raw}}$, so $\lambda \sim \varsigma e^{z_{\text{raw}}} \rightarrow 0$ and $\log \lambda$ keeps unit gradient however quiet the market gets.

The first three choices own simulation stability and calibratability; the fourth ensures the bound costs nothing in prediction when the market is quiet.

Rate-neutral mark head

Marks are decoded from the same h_t but never change the total rate:

$$p(e | t) = \text{softmax}(\text{MLP}(h_t))_e, \quad \lambda_e(t) = \lambda(t) p(e | t).$$

Because the soft-max lives on the simplex, the mark network can be arbitrarily deep — or later conditioned on exogenous state such as an agent’s resting quotes — without touching λ . This decouples the two failure modes: rate stability is owned entirely by the bounded head, mark expressiveness entirely by the simplex.

Likelihood and sampling

Training minimizes the negative log-likelihood of each training window, covering its events (t_i, e_i) and the next event after the window:

$$\mathcal{L} = \int_0^T \lambda(u | \mathcal{H}_u) du - \sum_i \left[\log \lambda(t_i) + \log p(e_i | t_i) \right],$$

with T the next-event time. The integral and the $\log \lambda$ terms score *when* events happen (the compensator charges for intensity placed where no event occurred); the $\log p$ term is a categorical cross-entropy scoring *what* each event is. By the factorization, the timing terms train only the bounded rate head and the mark term trains only the soft-max head. Each $\lambda(t_i)$ is evaluated at the left limit, the state just before event i , so an event never informs its own intensity.

The compensator integral is approximated by one intensity evaluation per inter-event interval (the endpoint rule); this under-weights a decaying intensity and biases the learned rate upward, which the post-hoc calibration below corrects. Closed-loop sampling uses Ogata thinning with the exact dominating rate $\varsigma \cdot \text{softplus}(z_{\max})$: propose arrivals from a homogeneous Poisson process at the ceiling and accept each with probability $\lambda(t) / (\varsigma \text{softplus}(z_{\max}))$.

Carried-state simulation

The standard roll-out protocol re-encodes a sliding window of the last W events from a zero state at every step. This is faithful to windowed training but has two costs: $O(W)$ work per simulated event, and a hard memory ceiling — nothing older than W events can influence the future, so long-memory stylized facts (the slow decay of $|r|$ autocorrelation) are *architecturally* unreachable regardless of the model. Because the backbone is a state-space recursion, both costs vanish: we carry the packed decoder state and advance it with only the new event, which is mathematically identical to encoding the entire history from the seed, at $O(1)$ cost per event — an order of magnitude faster than window re-encoding and memory-unbounded. Attention- and window-shaped baselines have no such Markov state and retain the sliding window.

Stateful (TBPTT) training

Windowed maximum likelihood encodes every training window from a cold initial state, so the model only ever trains in a “history just began” transient and MLE explains the observed rate with an inflated baseline — the free roll-out, whose state is warm, then double-counts. We train instead with truncated backpropagation through time: batch lanes walk contiguous stretches of the stream in order, and the decoder state at the end of window t is passed — *detached* — as the initial state of window $t+1$ (resets only at genuine stream boundaries, to the learned initial state). Gradients remain window-local; state values flow from the start of the stream. This makes the training regime consistent with carried-state simulation, at unchanged per-step cost; it requires non-overlapping windows (stride = window length), so our controls match that setting.

Post-hoc rate calibration

The factorization buys one more property: $\lambda = \varsigma \cdot \text{softplus}(z)$ is *exactly linear* in the learnable scale ς , the thinning ceiling $\varsigma \cdot \text{softplus}(z_{\max})$ scales with it, and the mark head is untouched by construction. The free-running rate — which is monotone but nonlinear in ς through the closed-loop feedback — can therefore be calibrated by a one-dimensional geometric bisection on a multiplier κ : roll out briefly at $\kappa \varsigma$, compare the realized rate to the target, and iterate over a handful of probe roll-outs. The result is a certificate-preserving, sampling-exact simulator at any prescribed mean rate. For *any* decoder, a constant κ multiplying the sim-time intensity is still mark-preserving — type probabilities λ_e / λ are unchanged — so the same bisection applies to every baseline; what is lost off-family is only the

exactness of the thinning bound. We make the procedure falsifiable: the target is the *validation*-split rate, the bisection must bracket and converge, and a full-scale validation roll-out must verify the target within tolerance (failures escalate to full-scale probes; unresolved failures exclude the checkpoint and are reported). Transfer to the test split is reported, never gated. We use calibration as a *post-hoc* correction for the residual rate inflation caused by the biased endpoint-rule training compensator — and, in the experiments, as a measurement instrument in its own right.

Experiments and Evaluations

We evaluate SS2P2 against five baselines on three questions: (1) does the bounded head cost predictive accuracy? (2) does SS2P2 reproduce the structure of real order flow in closed-loop simulation? (3) can each model be rate-calibrated at all? We close with computational efficiency.

Experimental Setup

Datasets. We benchmark on Coinbase BTC, ETH, and SOL event-driven order flow, seven days per asset, encoded by the event constructor into $|\mathcal{E}| = 62$ atomic types ($20 + 20 + 2 + 20$ over LO/CO/MO/IS, sides, and levels). Validation-split rates are 38.3, 47.9, and 24.0 events/s: production event rates, not subsampled streams.

Baselines. All models share the same input embedding and training configuration. They differ only in the decoder.

- **LSTM:** a standard LSTM decoder.
- **NHP** (Mei and Eisner 2017): the neural Hawkes continuous-time LSTM.
- **SAHP** (Zhang et al. 2020): a causal-attention decoder (adapted generic backbone).
- **PCT-LSTM** (Shi and Cartlidge 2022): the state-dependent parallel neural Hawkes process (CT4LSTM-PPP), ported faithfully; its market-state input is disabled because the closed-loop protocol provides no exogenous state stream.
- **S2P2** (Chang et al. 2025): the unbounded state-space parent; identical backbone to SS2P2, unbounded softplus head.

Training and evaluation protocol. Each model trains per asset with sequence length 1024, non-overlapping windows, identical gradient-step counts, and three training seeds. The fixed 1024-event context spans only ~ 20 –40 s of market time at these rates, so long-memory structure leans entirely on carried state. Evaluation is strict end-to-end:

- **Prediction** is scored on *every* genuine test event by a lossless streaming evaluator with carried state (one cold start per genuine segment boundary); the evaluator hard-fails if any event goes unscored.
- **Simulation** uses carried-state closed-loop roll-outs (32×600 s, three roll-out seeds per checkpoint) against equal-duration bootstrap samples of real segments. SS2P2 samples by Ogata thinning with its exact closed-form ceiling. Baselines, which admit no exact bound, share a common compensator-inversion sampler.

- **Calibration:** every model is rate-calibrated before scoring by a one-dimensional bisection on the time-rescaling constant κ , targeting the *validation*-split rate. A full-scale validation roll-out must then verify the target within 15%. Checkpoints that fail are excluded and the failure is reported as a result.
- **Statistics:** roll-out seeds are averaged within each checkpoint. 95% CIs are *t*-based across checkpoints. Any batch failure or non-finite value aborts the run.

Prediction Quality

SS2P2 is the strongest predictor on two of the three assets and bounding the intensity does not cost predictive accuracy. According to Table 1, SS2P2 wins NLL and accuracy on SOL by a wide margin (0.27 vs. the strongest baseline’s 0.38 overall NLL; 0.192 vs. 0.184 accuracy), wins both outright on ETH (-0.87 vs. PCT-LSTM’s -0.84 ; 0.345 vs. NHP’s 0.341), and on BTC ties NHP on marks (0.448 vs. 0.449) while conceding overall likelihood. Expected-time MAE tells the same story from the metric side: SS2P2 predicts the next event time within 0.030–0.050 s of the truth on every asset, within about a millisecond of the best model, while the unbounded S2P2 is pathological (2.3, 10.2, 46.2 s): its intensity-collapse states place essentially no mass where events occur, and the expectation integral diverges toward the horizon. SS2P2 achieves all of this while remaining verifiably calibratable in closed loop (Table 2); no baseline combines both.

The remaining gap to NHP is in timing on BTC, and a per-gap decomposition of $\text{timeNLL} = -\log \lambda(t_i) + \Delta u_i$ (event term + compensator mass) shows it is not about bursts: the state-space family’s event terms beat NHP’s (-4.87 SS2P2 vs. -4.29). The entire gap is compensator mass in *quiet gaps*: mean per-gap rescaled mass is 1.85 (SS2P2) and 2.38 (S2P2) against NHP’s near-ideal 1.07 (target 1), with the excess accruing in gaps beyond 0.1 s: SS2P2 integrates 4.3 units of intensity across a 0.5–2 s lull where NHP integrates 0.8. The insight is mechanistic: NHP decays toward a *learned steady state* that tracks the falling empirical hazard of long gaps, while the state-space family, with no dedicated resting-rate parameter, stays hot through lulls. This quiet-gap cost is the deliberate price of stability; what it buys is the subject of the next two subsections.

Calibratability

Post-hoc rate calibration applies to *every* decoder, because a constant κ rescales sim-time intensity without touching type probabilities, so we calibrate every model and treat the *outcome* as an experimental result. The protocol is deliberately falsifiable: probes must bracket and converge to the validation target, a full-scale roll-out must verify it, and every failure is reproducible (probe seeds are deterministic; each re-tried failure reproduced its bisection bracket bit-identically).

Whether a model can be calibrated at all is a measurable property, and the bound decides it. Table 2 is, to our knowledge, the first cross-asset quantification of closed-loop calibratability for neural TPPs: the bounded SS2P2 head verifies on *all nine* checkpoints, while the unbounded S2P2 head

model	BTC			ETH			SOL		
	NLL↓	ACC↑	MAE↓	NLL↓	ACC↑	MAE↓	NLL↓	ACC↑	MAE↓
NHP	-1.07±0.01	0.449±0.003	0.029±0.000	-0.81±0.03	0.341±0.006	0.030±0.000	0.53±0.01	0.184±0.004	0.052±0.001
LSTM	-0.44±0.40	0.411±0.015	0.032±0.002	0.15±0.04	0.288±0.003	0.034±0.002	1.34±0.25	0.141±0.005	0.057±0.004
SAHP†	-0.93±0.01	0.412±0.001	0.034±0.005	-0.60±0.01	0.297±0.004	0.038±0.006	0.67±0.02	0.155±0.003	0.060±0.002
PCT-LSTM	-0.97±0.03	0.380±0.009	0.029±0.000	-0.84±0.05	0.272±0.014	0.028±0.000	0.38±0.08	0.130±0.011	0.048±0.000
S2P2	-0.39±0.25	0.439±0.002	2.3±5.9	-0.38±0.22	0.331±0.008	10.2±31.0	0.67±0.49	0.182±0.002	46.2±96.0
SS2P2	-0.87±0.12	0.448±0.001	0.030±0.001	-0.87±0.08	0.345±0.003	0.031±0.001	0.27±0.05	0.192±0.001	0.050±0.001

Table 1: Prediction per genuine test event (mean $\pm$ 95% CI over three training seeds; streaming lossless evaluator). NLL = overall negative log-likelihood (time + mark); ACC = next-type accuracy; MAE = expected-time mean absolute error in seconds. SS2P2 wins NLL and ACC on ETH and SOL and ties NHP on BTC accuracy; S2P2’s MAE column is the intensity-collapse tail of the unbounded head.

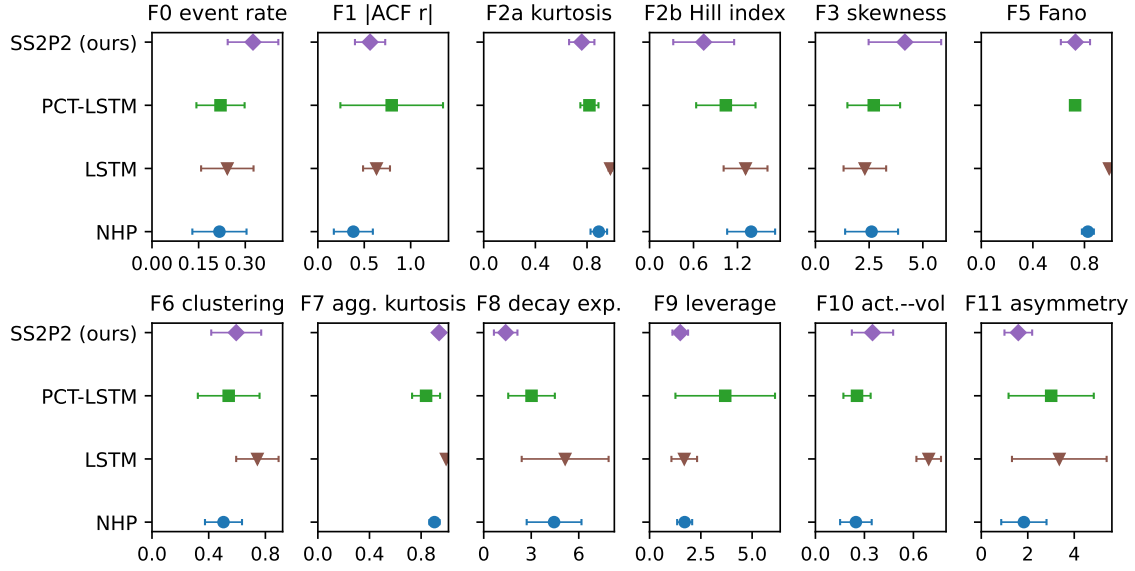


Figure 3: The full stylized-fact battery, pooled across the three assets: relative error vs. real for event rate (F0), Fano factor averaged over 1–50 s scales (F5), and the ten distributional facts F1–F11 (whiskers = 95% CI over asset-checkpoint pairs; roll-out seeds averaged within checkpoint). S2P2 and SAHP omitted: too few calibrated checkpoints.

model	BTC	ETH	SOL
	38 ev/s	48 ev/s	24 ev/s
NHP	3/3	3/3	3/3
LSTM	3/3	3/3	3/3
SAHP†	<i>model-level divergence: reported uncalibrated</i>		
PCT-LSTM	3/3	3/3	3/3
S2P2	2/3	1/3	0/3
SS2P2	3/3	3/3	3/3

Table 2: Checkpoints passing calibration + full-scale verification, out of three training seeds per asset (bold marks failures). The unbounded S2P2 head fails on 7 of 9 checkpoints, including *all three* on SOL where no usable simulator remains, while the bounded SS2P2 head verifies on all nine. SAHP diverges at the model level on every asset.

fails on 7 of 9, and on SOL the model class is excluded outright.

Figure 4 shows the failure in detail. The closed-loop rate of a failing S2P2 checkpoint is near-discontinuous in κ : an 8×10^{-4} change flips the realized rate across the entire tolerance band ($34.8 \rightarrow 42.3$ ev/s), and repeated probes at fixed κ differ by $\sim 20\%$. This is the behaviour of a system parked near criticality, where no constant time-rescaling exists. The SS2P2 response crosses the same target smoothly on every asset. (An earlier training run of one BTC seed produced a checkpoint whose single failure was qualitatively different: a per-lane diagnostic traced it to a *metastable* burst-locked regime at $\sim 2\times$ the target rate that individual roll-out lanes enter stochastically, bounded and benign next to S2P2’s criticality; retraining escaped that basin.)

The insight closes the loop on the certificate: the same unbounded softplus head whose intensity-collapse states produce S2P2’s pathological expected-time tail (Table 1) also

places checkpoints near criticality in closed loop, and both problems disappear under the softmin bound.

Simulation Quality

SS2P2 is the only model that pairs competitive closed-loop simulation with top-tier prediction. Figure 3 scores the calibrated roll-outs on the full stylized-fact battery (event rate, Fano factor, and the ten distributional facts), pooled across the three assets: SS2P2 sits at or near the front on most facts and is clearly strongest on aggregated kurtosis (F7), the volatility decay exponent (F8), and heavy tails (F2a); every model’s rate is separately verified against its target during calibration, so these are structural comparisons at matched clocks.

The per-asset Fano view in Figure 5 details the burst structure. PCT-LSTM and SS2P2 track the rising real curve furthest on every asset, with PCT-LSTM slightly ahead on BTC and ETH and SS2P2 ahead on SOL; but PCT-LSTM has the weakest type accuracy in Table 1, so its simulation strength does not come with prediction strength. NHP, the best BTC predictor, flattens by ~ 5 s on every asset; the LSTM simulates a near-Poisson stream (its Fano curve stays flat near 1), the trivially-stable end of the trade-off; and the one surviving S2P2 arm on ETH calibrates in rate but over-disperses wildly, so rate calibration cannot repair a near-critical checkpoint’s structure. Real Fano rises from ~ 70 at 1 s to ~ 1300 at 50 s; every model under-disperses at this context length, SS2P2 and PCT-LSTM least.

The insight is that prediction and simulation quality are distinct axes: no unbounded model wins both, and the two families that simulate burst structure best do it through opposite designs (per-type parallel streams vs. a bounded shared rate), of which only the bounded design also predicts at the top of the table.

Efficiency Analysis

The state-space backbone is also computationally advantageous once its per-event sequential loop is replaced by an exact associative prefix scan (Hillis–Steele over the ZOH recurrence; identical states and gradients to 2×10^{-4}). Table 3 reports wall-clock measurements on one GPU (ETH checkpoints, benchmark configuration). The scan collapses the S2P2-family training step from 3.5 s to ~ 0.06 s: attention-class speed, $60\times$ faster than the loop and $73\times$ faster than the CT-LSTM, turning a multi-hour training run into minutes. For sampling, SS2P2’s exact ceiling makes Ogata thinning available with a *constant* dominating rate (no per-step bound estimation; mean acceptance 0.54 on ETH): exactness costs $\sim 2.5\times$ throughput against the approximate inversion sampler shared by all models, and both run far above real time (353 vs. 48 events/s on the fastest asset).

Discussion and Conclusion

We held six neural MTPPs to both sides of the prediction–simulation tension on three Coinbase assets at production event rates (24–48 events/s), under a protocol in which failures are visible: lossless streaming prediction, multi-seed

	train step (s)	vs. seq. loop	sim. ev/s
NHP	4.35	–	1253
LSTM	0.019	–	2083
SAHP	0.059	–	583
PCT-LSTM	0.79	–	1919
S2P2	0.055	$64\times$	1163
SS2P2	0.059	$60\times$	879 / 353 [†]

Table 3: Efficiency on one GPU. Training: one fwd+bwd on 64×1024 events (scan for the S2P2 family; speedup vs. their sequential loop). Simulation: events per wall-second, 8×60 s roll-outs. [†]inversion / exact-ceiling Ogata thinning.

confidence intervals, and verified rate calibration whose failures count as results. The SS2P2 heads win every prediction metric on SOL, lead ETH, tie the strongest baseline on BTC marks, and remain verifiably calibratable with competitive simulated structure throughout. The sharpest finding is that *calibratability is a model property*: the unbounded parent head fails verified rate calibration on 7 of 9 checkpoints, including all three on SOL, with a critical-point signature, while the bounded head verifies on all nine, confirming the stability certificate empirically. Because the mark head is rate-neutral, conditioning it on exogenous state such as an agent’s resting quotes cannot destabilize the rate, so SS2P2 can serve as an environment model for market-making world models. That action-conditioned extension is future work. The same factorized intensity supports the importance-sampling queries of Chang, Boyd, and Smyth (2024), which map onto quantities a market maker needs: fill probability and adverse-selection risk. All tables and figures are regenerated by the released scripts.

References

- Byrd, D.; Hybinette, M.; and Balch, T. H. 2020. ABIDES: Towards High-Fidelity Multi-Agent Market Simulation. In *Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation*, 11–22.
- Chang, Y.; Boyd, A.; and Smyth, P. 2024. Probabilistic Modeling for Sequences of Sets in Continuous-Time. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 238 of *PMLR*.
- Chang, Y.; Boyd, A.; Xiao, C.; Kass-Hout, T.; Bhatia, P.; Smyth, P.; and Warrington, A. 2025. Deep Continuous-Time State-Space Models for Marked Event Sequences. In *Advances in Neural Information Processing Systems 38*. ArXiv:2412.19634.
- Cloudburst. 2025. Cloudburst Technologies.
- Coletta, A.; Moulin, A.; Vyetenko, S.; and Balch, T. 2022. Learning to Simulate Realistic Limit Order Book Markets from Data as a World Agent. In *Proceedings of the 3rd ACM International Conference on AI in Finance*.
- Coletta, A.; Prata, M.; Conti, M.; Mercanti, E.; Bartolini, N.; Moulin, A.; Vyetenko, S.; and Balch, T. 2021. Towards Realistic Market Simulations: A Generative Adversarial Networks Approach. In *Proceedings of the Second ACM International Conference on AI in Finance*.

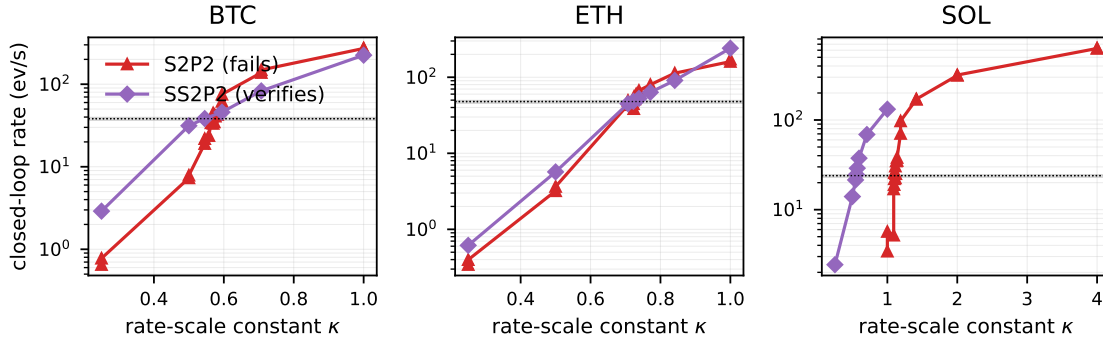


Figure 4: The failing S2P2 checkpoints’ closed-loop rates are near-discontinuous in the rescaling constant κ on every asset, most extremely on SOL, while the verified SS2P2 checkpoints cross each target smoothly. Curves are calibration probe ladders (seed-1 checkpoints; target band shaded).

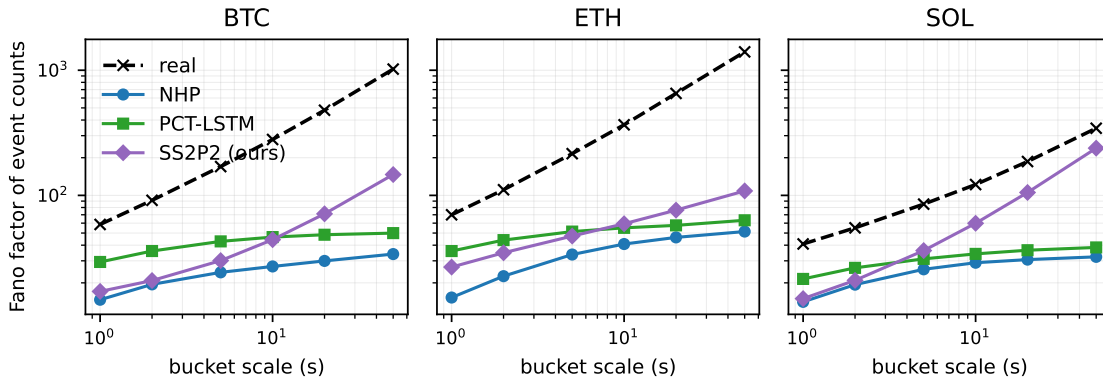


Figure 5: Real order flow is super-Poissonian with Fano *rising* in scale, and every model under-disperses at this context length, SS2P2 and PCT-LSTM least; the pattern is identical on all three assets. Curves are Fano factors of simulated event counts vs. bucket scale (mean over calibrated checkpoints and roll-out seeds).

Cont, R. 2001. Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues. *Quantitative Finance*, 1(2): 223–236.

Du, N.; Dai, H.; Trivedi, R.; Upadhyay, U.; Gomez-Rodriguez, M.; and Song, L. 2016. Recurrent Marked Temporal Point Processes: Embedding Event History to Vector. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1555–1564.

Frey, S.; Li, K.; Nagy, P.; Sapora, S.; Lu, C.; Zohren, S.; Foerster, J.; and Calinescu, A. 2023. JAX-LOB: A GPU-Accelerated Limit Order Book Simulator to Unlock Large-Scale Reinforcement Learning for Trading. In *Proceedings of the 4th ACM International Conference on AI in Finance*.

Hawkes, A. G. 1971. Spectra of Some Self-Exciting and Mutually Exciting Point Processes. *Biometrika*, 58(1): 83–90.

Hawkes, A. G. 2018. Hawkes processes and their applications to finance: a review. *Quantitative Finance*, 18(2): 193–198.

Huang, Y.-H.; Xu, C.; Liu, Y.; Liu, W.; Li, W.-J.; and Bian, J. 2026. Controllable Financial Market Generation with Diffusion Guided Meta Agent. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence*.

Jain, K.; Firoozye, N.; Kochems, J.; and Treleaven, P. 2024a. Limit Order Book dynamics and order size modelling using Compound Hawkes Process. *Finance Research Letters*, 69: 106157.

Jain, K.; Firoozye, N.; Kochems, J.; and Treleaven, P. 2024b. Limit Order Book Simulations: A Review. *arXiv preprint arXiv:2402.17359*.

Kaiko. 2025. The crypto industry’s leading data, analytics, and indices provider. - Kaiko.

Li, J.; Liu, Y.; Liu, W.; Fang, S.; Wang, L.; Xu, C.; and Bian, J. 2025. MarS: A Financial Market Simulation Engine Powered by Generative Foundation Model. In *International Conference on Learning Representations*.

Li, J.; Wang, X.; Lin, Y.; Sinha, A.; and Wellman, M. P. 2020. Generating Realistic Stock Market Order Streams.

In *Proceedings of the 34th AAAI Conference on Artificial Intelligence*, 727–734.

Mei, H.; and Eisner, J. M. 2017. The Neural Hawkes Process: A Neurally Self-Modulating Multivariate Point Process. In Guyon, I.; Luxburg, U. V.; Bengio, S.; Wallach, H.; Fergus, R.; Vishwanathan, S.; and Garnett, R., eds., *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

Morariu-Patrichi, M.; and Pakkanen, M. S. 2022. State-dependent Hawkes processes and their application to limit order book modelling. *Quantitative Finance*, 22(3): 563–583.

Nagy, P.; Frey, S.; Li, K.; Sarkar, B.; Vyetrenko, S.; Zohren, S.; Calinescu, A.; and Foerster, J. 2025. LOB-Bench: Benchmarking Generative AI for Finance – an Application to Limit Order Book Data. In *Proceedings of the 42nd International Conference on Machine Learning*.

Nagy, P.; Frey, S.; Sapora, S.; Li, K.; Calinescu, A.; Zohren, S.; and Foerster, J. 2023. Generative AI for End-to-End Limit Order Book Modelling: A Token-Level Autoregressive Generative Model of Message Flow Using a Deep State Space Network. In *Proceedings of the 4th ACM International Conference on AI in Finance*.

Shchur, O.; Türkmen, A. C.; Januschowski, T.; and Günnemann, S. 2021. Neural Temporal Point Processes: A Review. *IJCAI International Joint Conference on Artificial Intelligence*, 4585–4593.

Shi, Z.; and Cartlidge, J. 2022. State Dependent Parallel Neural Hawkes Process for Limit Order Book Event Stream Prediction and Simulation. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 1607–1615. New York, NY, USA: ACM. ISBN 9781450393850.

Sood, S.; Kawawa-Beaudan, J.; Borrajo, D.; and Veloso, M. 2026. TradeFM: A Generative Foundation Model for Trade-flow and Market Microstructure. *arXiv preprint arXiv:2602.23784*.

Vyetrenko, S.; Byrd, D.; Petosa, N.; Mahfouz, M.; Dervovic, D.; Veloso, M.; and Balch, T. 2020. Get Real: Realism Metrics for Robust Limit Order Book Market Simulations. In *Proceedings of the First ACM International Conference on AI in Finance*.

Zhang, Q.; Lipani, A.; Kirnap, O.; and Yilmaz, E. 2020. Self-Attentive Hawkes Process. In *International Conference on Machine Learning*, 11183–11193.

Zhang, S.; Zhou, C.; Liu, Y.; Zhang, P.; Lin, X.; and Ma, Z.-M. 2024. Neural Jump-Diffusion Temporal Point Processes. In *Proceedings of the 41st International Conference on Machine Learning*.

Zuo, S.; Jiang, H.; Li, Z.; Zhao, T.; and Zha, H. 2020. Transformer Hawkes Process. In *Proceedings of the 37th International Conference on Machine Learning*.